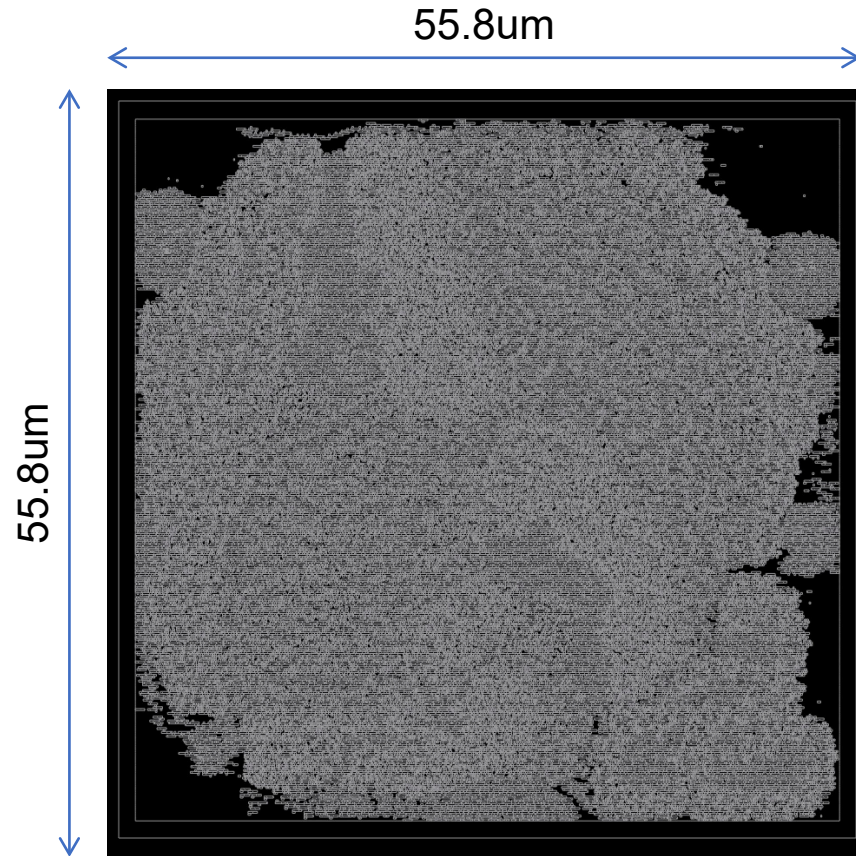
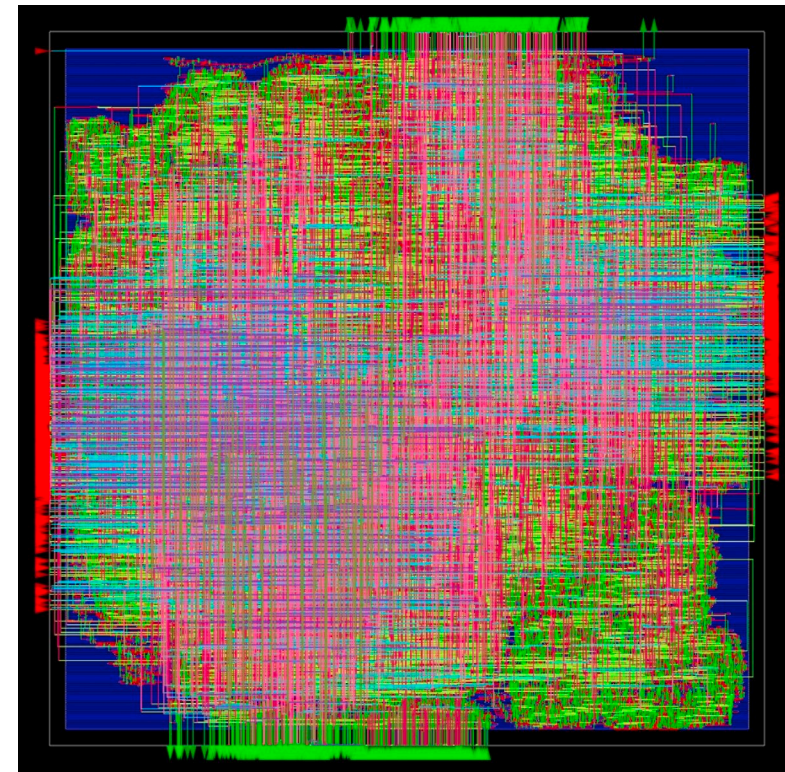


Benchmark: ECG (62K Gates)

PDK: USC-3N-2D (Front-side BEOL), EDA: OpenROAD



Placement



Routing

Benchmark: ECG (62K Gates)

PDK: USC-3N-2D (Front-side BEOL), EDA: OpenROAD

- Intended for PDK+EDA flow validation, not PPA optimization.

Performance	Target freq (GHz)	2.00
	Effective freq (GHz)	1.99
	WNS (ps)	-1.30
	TNS (ps)	-23.97
Power	Total power (mW)	22.01
	Internal power (mW)	16.64
	Switching power (mW)	5.35
	Leakage power (mW)	0.03
Area	Chip dimension (um)	55.8 x 55.8
	Macro Count	0
	Cell Count	61,585
	Placement Density	72%

Benchmark: ECG (62K Gates)

PDK: USC-3N-2D (Front-side BEOL), EDA: OpenROAD

- Intended for PDK+EDA flow validation, not PPA optimization.

Power delivery network	PDN WL (um)	19,720
	PDN stripe width (um)	0.012
	PDN stripe pitch (um)	0.144
Clock delivery network	Clock net count	310
	Clock WL (um)	5,037
	Clock power (mw)	4.7

Placement (min)	0.63
CTS (min)	0.12
Routing (min)	13.9
Total (min)	14.7

Turnaround time

OS	Red Hat 8.10
CPU	Xeon Gold 6454S
#Cores	64
CPU Speed	3400 MHz
Memory	512 GB

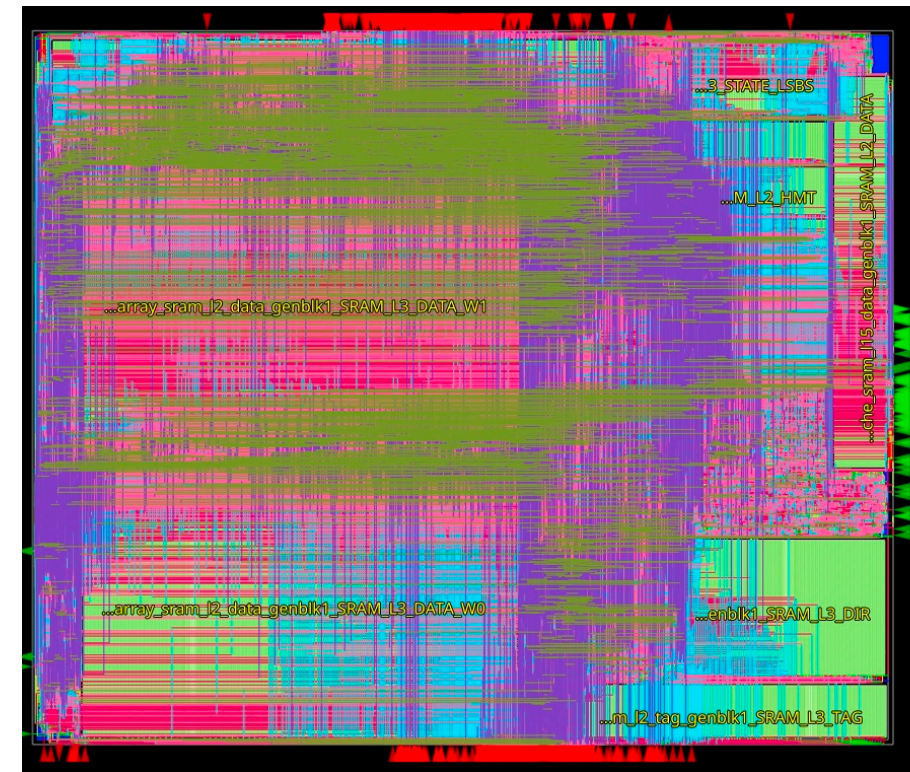
Server spec

Benchmark: OpenPiton (292K Gates)

PDK: USC-3N-2D (Front-side BEOL), EDA: OpenROAD



Placement



Routing

Benchmark: OpenPiton (292K Gates)

PDK: USC-3N-2D (Front-side BEOL), EDA: OpenROAD

- Intended for PDK+EDA flow validation, not PPA optimization.

Performance	Target freq (GHz)	1.00
	Effective freq (GHz)	0.95
	WNS (ps)	-54.34
	TNS (ps)	-258.07
Power	Total power (mW)	101.87
	Internal power (mW)	94.75
	Switching power (mW)	6.39
	Leakage power (mW)	0.72
Area	Chip dimension (um)	330.0 x 398.2
	Macro Count	28
	Cell Count	292,047
	Placement Density	72%

Benchmark: OpenPiton (292K Gates)

PDK: USC-3N-2D (Front-side BEOL), EDA: OpenROAD

- Intended for PDK+EDA flow validation, not PPA optimization.

Power delivery network	PDN WL (um)	283,598
	PDN stripe width (um)	0.012
	PDN stripe pitch (um)	0.144
Clock delivery network	Clock net count	16,811
	Clock WL (um)	241,783
	Clock Power (mw)	11.2

Placement (min)	9.5
CTS (min)	2.1
Routing (min)	32.4
Total (min)	43.9

Turnaround time

OS	Red Hat 8.10
CPU	Xeon Gold 6454S
#Cores	64
CPU Speed	3400 MHz
Memory	512 GB

Server spec